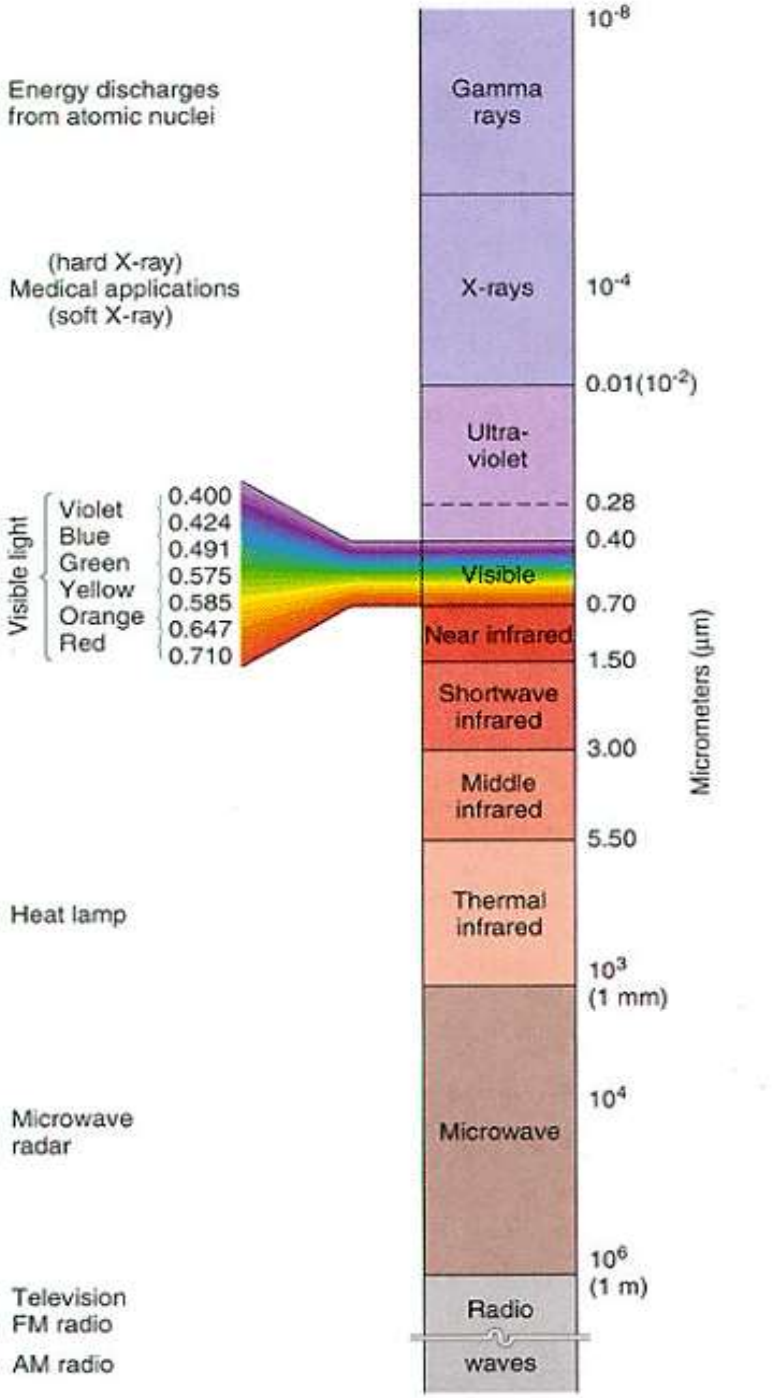
Lesson 4:
COLOUR

Physical Background

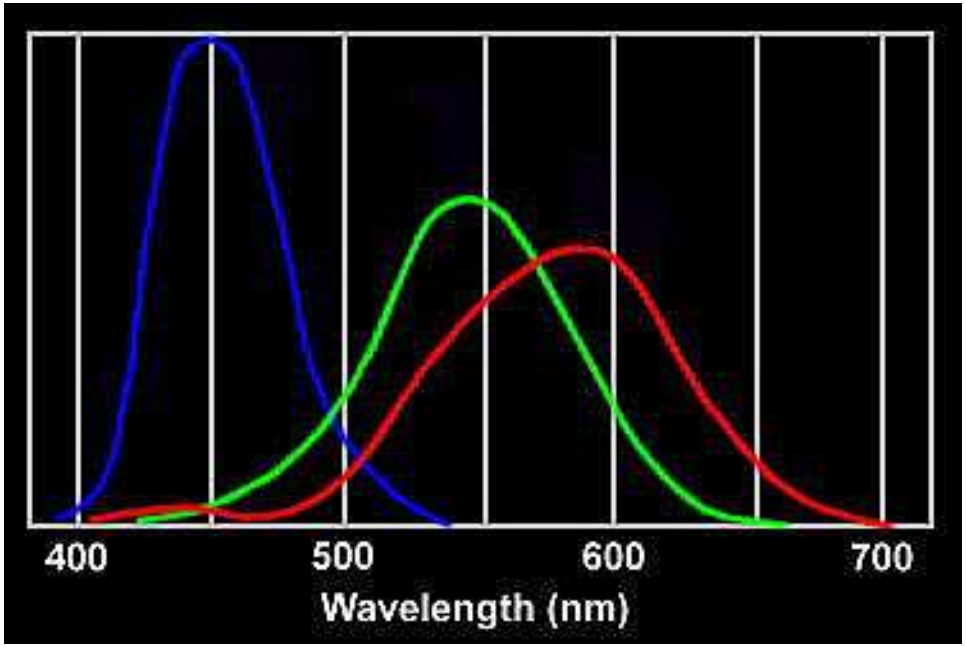
- **Visible light**: a narrow band of electromagnetic radiation → **380nm (blue)** – **780nm (red)**
- **Wavelength**: Each physically distinct colour corresponds to at least one wavelength in this band.
- **Spectrum**: Intensity as a function of wavelength
- **The colour of an object**: is the product of the spectrum of the incident light with the light absorption and/or reflection properties of the object.



source: Christopherson (2000) *Geosystems*

Human Colour perception

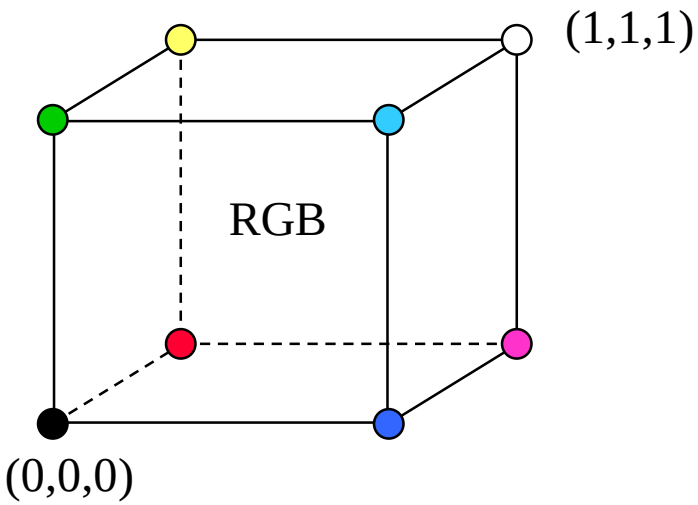
- The human eye does not perceive individual light wavelengths.
- It contains three types of colour receptor (cones) which integrate over parts of the spectrum:



From http://math.ucr.edu/home/baez/physics/General/BlueSky/blue_sky.html

Human Colour perception

- It is therefore possible to characterise a psycho-visual colour by specifying the amounts of three primary colours: red, green and blue, mixed together.
- This leads to the standard RGB space used in television, computer monitors, etc.
- We specify the levels of R, G and B in the range [0, 1], but they can easily be extended to other ranges (8-bit integers for example).



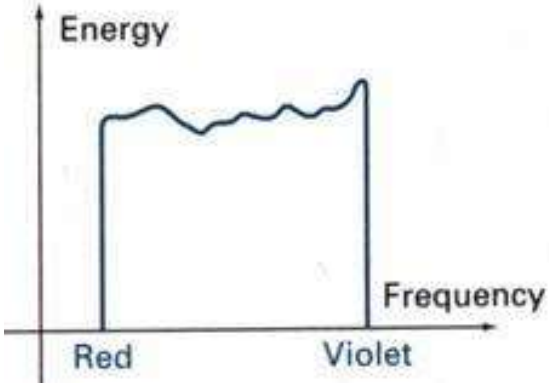
Intuitive Colour Concepts

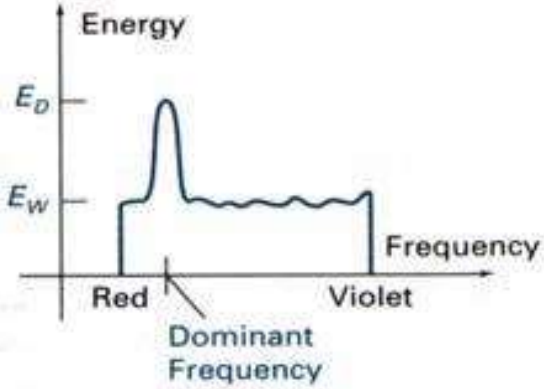
- Colour mixing created by an artist
- Shades, tints and tones in scene can be produced by mixing colour pigments (hues) with white and black pigments

- Shades
  - Add black pigment to pure colour
  - The more black pigment, the darker the shade
- Tints
  - Add white pigment to the original colour
  - Making it lighter as more white is added
- Tones
  - Produced by adding both black and white pigments

Characteristics of Colour

- Dominant frequency (hue, colour)
- Brightness (area under the curve), total light energy
- Purity (saturation), how close a light appear to be a pure spectral colour, such as red
- Chromaticity is used to refer collectively to the two properties describing colour characteristics: purity and dominant frequency





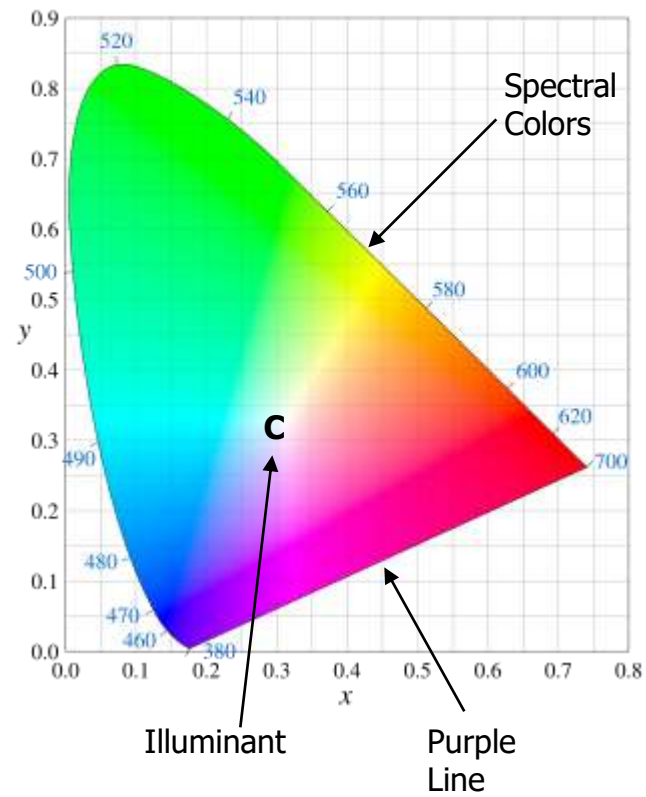
The CIE Chromaticity Diagram

- CIE: International Commission on Illumination (Comission Internationale de l'Eclairage).
- Shows colour compositions as a function of x(**red**) and y(**green**)
- Luminance values are not available because of normalization
- Colors with different luminance but same chromaticity map to the same point

- Usage of CIE chromaticity diagram
  - Comparing colour gamuts for different set of primaries
  - Identifying complementary colors
  - Determining purity and dominant wavelength for a given colour

The CIE Chromaticity Diagram

- A tongue-shape curve formed by plotting the normalized amounts x and y for colours in the visible spectrum
  - Points along the curve are spectral colour (pure colour)
  - Purple line, the line joining the red and violet spectral points
  - Illuminant C, plotted for a white light source and used as a standard approximation for average daylight



The CIE Chromaticity Diagram

- To determine the range of colors that can be obtained from the 3 given colors in the Chromaticity Diagram, we simply draw connecting lines to each of the three colour points.
- The result is a triangle and any colour inside a triangle is produced by various combinations of the three initial colours.
- The triangle shows a typical range of colours (called the colour gamut) produced by RGB monitor

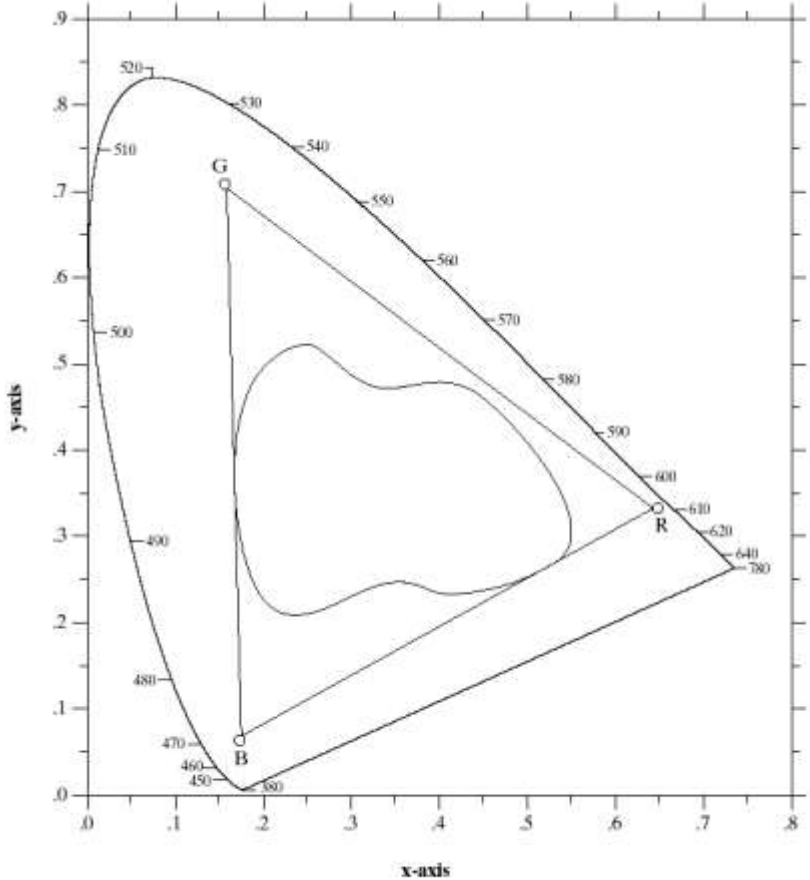


FIGURE 6.6 Typical color gamut of color monitors (triangle) and color printing devices (irregular region).

Complementary Colours

- Additive
  - Blue is one-third
  - Yellow (red+green) is two-thirds
  - When blue and yellow light are added together, they produce white light

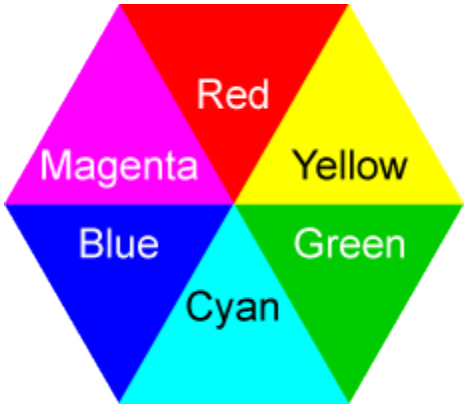
- Pair of complementary colours
  - blue and yellow
  - green and magenta
  - red and cyan

addRG

subCR

- Subtractive
  - Orange (between red and yellow)<>cyan-blue
  - green-cyan<>magenta-red colour

subYM



The CIE Chromaticity Diagram

- Complementary colors
  - Represented on the diagram as two points on opposite sides of C and collinear with C
  - The distance of the two colors C1 and C2 to C determine the amount of each needed to produce white light

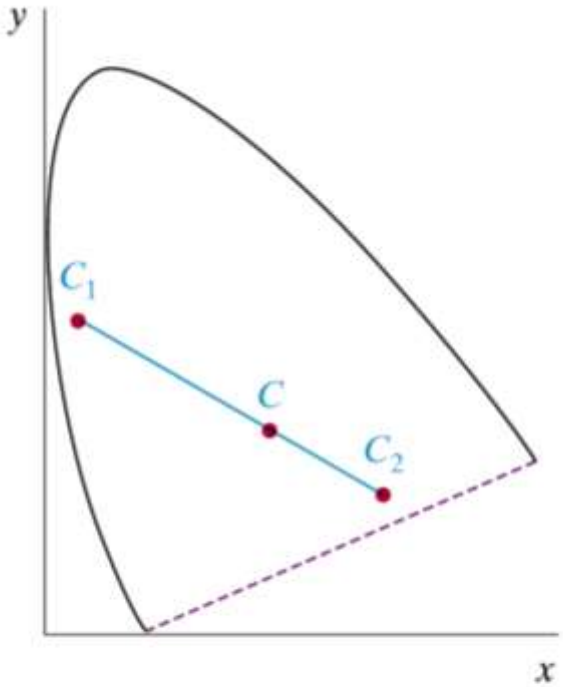


Figure 12-9

Representing complementary colors on the chromaticity diagram.

The CIE Chromaticity Diagram

- Dominant wavelength
  - Draw a straight from C through colour point to a spectral colour on the curve, the spectral colour is the dominant wavelength
  - Special case: a point between C and a point on the purple line C_p, take the compliment C_{sp} as dominant
- Purity
  - For a point C1, the purity determined as the relative distance of C1 from C along the straight line joining C to Cs
  - Purity ratio = d_{C1} / d_{Cs}

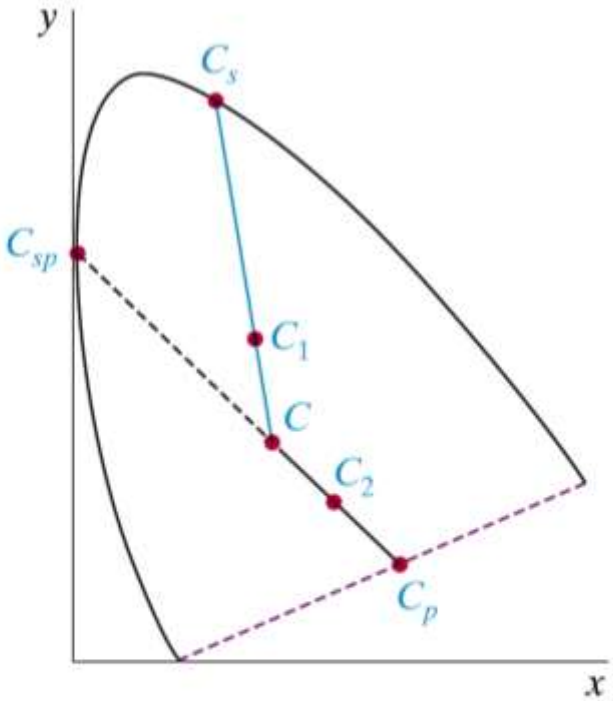


Figure 12-10

Determining dominant wavelength and purity using the chromaticity diagram.

Colour Models

- Method for explaining the properties or behavior of colour within some particular context
  - Combine the light from two or more sources with different dominant frequencies and vary the intensity of light to generate a range of additional colors
- Primary Colors
  - 3 primaries are sufficient for most purposes
- **Colour gamut** is the set of all colors that we can produce from the primary colors
- **A Complementary colour** is two primary colors that produce white
  - Red and Cyan, Green and Magenta, Blue and Yellow

Colour Models

- The purpose of a colour model (also called colour Space or colour System) is to facilitate the specification of colors in some standard way

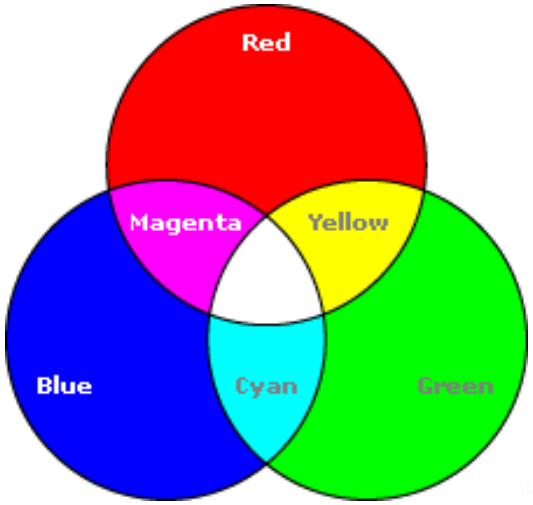
- A colour model is a specification of a coordinate system and a subspace within that system where each colour is represented by a single point

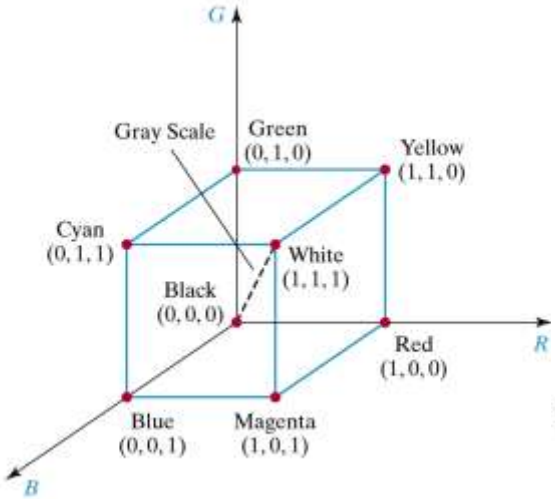
- **Colour Models**

RGB (Red, Green, Blue) - used with color CRT monitors
CMY (Cyan, Magenta, Yellow)- used for colour printing
CMYK (Cyan, Magenta, Yellow, Black) - colour printing
HSI (Hue, Saturation, Intensity)
YIQ (Luminance,In phase, Quadrature) - broadcast TV color system

The RGB Colour Model

- Basic theory of RGB colour model
  - The tristimulus theory of vision: It states that human eyes perceive colour through the stimulation of three visual pigment of the cones of the retina
    - Red, Green and Blue
  - Model can be represented by the unit cube defined on R,G and B axes





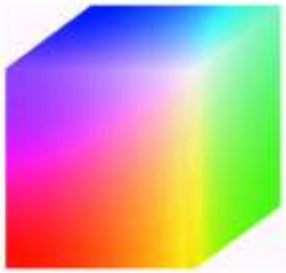


FIGURE 6.8 RGB 24-bit color cube.

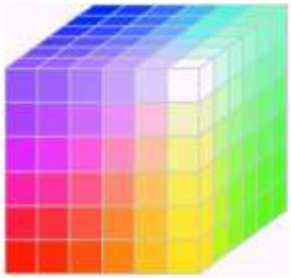


FIGURE 6.11 The RGB safe-color cube.

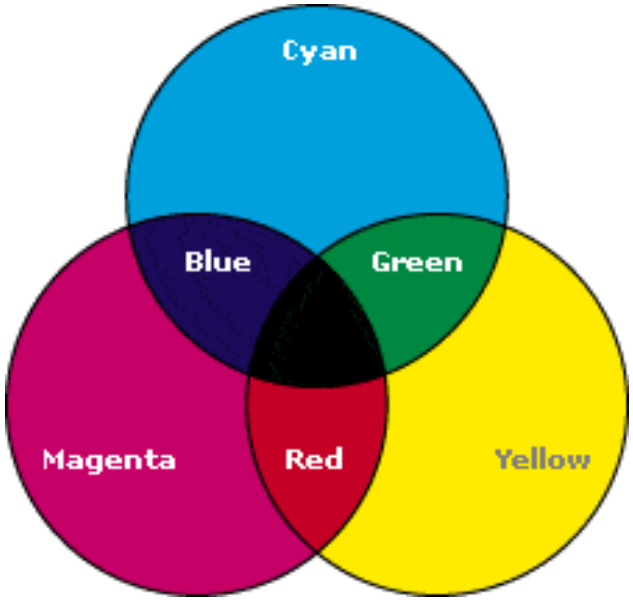
The RGB Colour Model

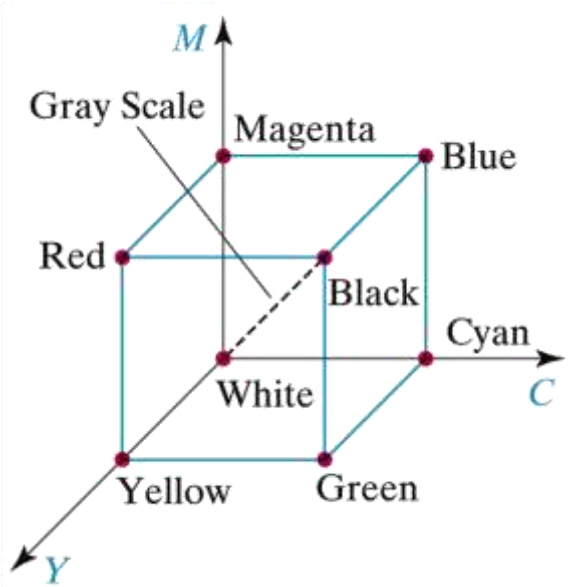
- The primary colors are red, green, and blue.
- It is an additive model, in which colors are produced by adding components, with white having all colors present and black being the absence of any colour.
- Used for active displays such as television and computer screens.

- The RGB model is usually represented by a unit cube with one corner located at the origin of a three-dimensional colour coordinate system, the axes being labeled R, G, B, and having a range of values [0, 1].
- The origin (0, 0, 0) is considered black and the diagonally opposite corner (1, 1, 1) is called white. The line joining black to white represents a gray scale and has equal components of R, G, B.

The CMY and CMYK Colour Models

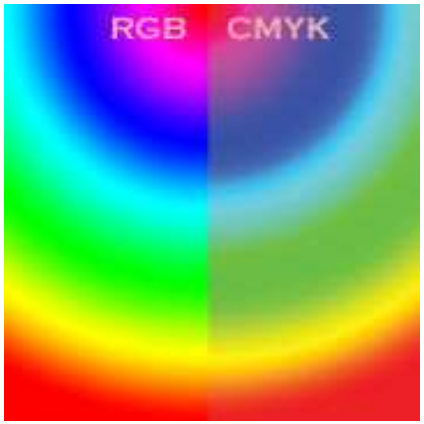
- Colour models for hard-copy devices, such as printers
  - Produce a colour picture by coating a paper with colour pigments
  - Obtain colour patterns on the paper by reflected light, which is a subtractive process
- The CMY parameters
  - A subtractive colour model can be formed with the primary colors cyan, magenta and yellow
  - Unit cube representation for the CMY model with white at origin





The CMY and CMYK Colour Models

- Most devices that deposit colour pigments on paper (such as colour Printers and Copiers) requires CMY data input or perform RGB to CMY conversion internally



- Transformation between RGB and CMY colour spaces
  - Transformation matrix of conversion from RGB to CMY

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

  - Transformation matrix of conversion from CMY to RGB

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix}$$

CMY and CMYK Colour Model

- CMY is a Subtractive colour Model, where adding different pigments causes various colors not to be reflected and thus not to be seen.
- Here, white is the absence of colors, and black is the sum of all of them
- Equal amounts of Pigment primaries (Cyan, Magenta and Yellow) should produce Black
- In practice combining these colors for printing produces a "Muddy-Black" colour
- So in order to produce "True-Black" a fourth colour "Black" is added giving rise to CMYK model

HSI colour Model

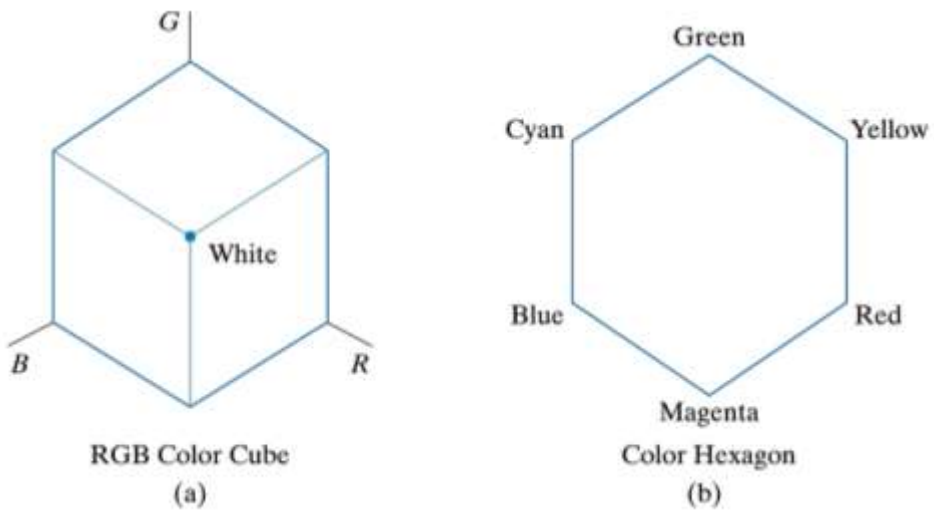
- Hue (dominant colour seen)
  - Wavelength of the pure colour observed in the signal.
  - Distinguishes red, yellow, green, etc.
  - More the 400 hues can be seen by the human eye.
- Saturation (degree of dilution)
  - Inverse of the quantity of "white" present in the signal. A pure colour has 100% saturation, the white and grey have 0% saturation.
  - Distinguishes red from pink, marine blue from royal blue, etc.
  - About 20 saturation levels are visible per hue.
- Intensity
  - Distinguishes the gray levels.
- Two values (Hue & Saturation) encode *chromaticity*
- Intensity encode *monochrome part.*

The YIQ and Related colour Models

- A recoding of RGB for transmission efficiency and for downward compatibility with black and white television.
- Here Y=luminance, the same as the CIE Y primary. Only the Y component of a color TV signal is shown on black-and-white TVs.
- Chromaticity information (hue and purity) is incorporated into the I and Q parameters
- I stands for in-phase, while Q stands for quadrature, referring to the components used in quadrature amplitude modulation.
- Transmitted using NTSC (National Television System Committee) standard
- More bits of bandwidth are used to represent Y than to represent I and Q, because our eye is more sensitive to changes in luminance

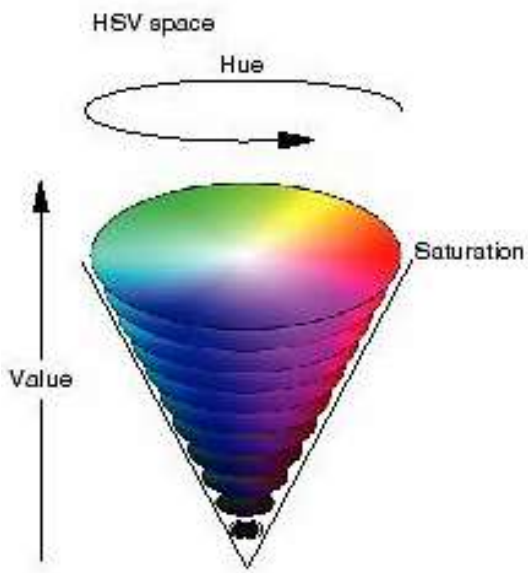
The HSV colour Model

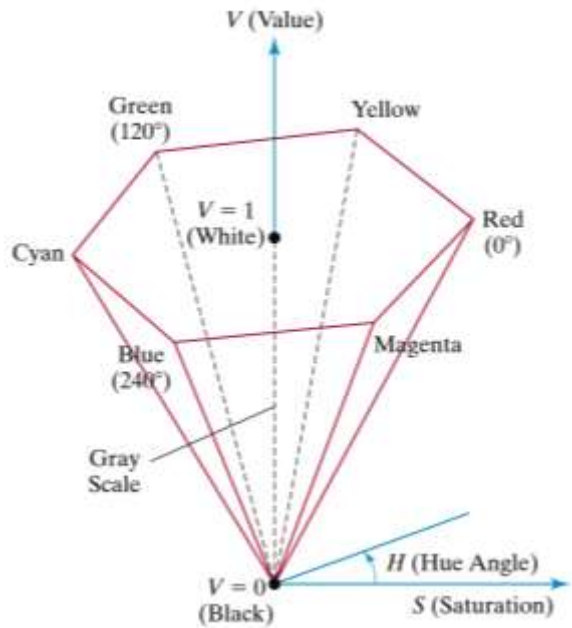
- The HSV parameters
  - Colour parameters are hue (H), saturation (S) and value (V)
  - Derived by relating the HSV parameters to the direction in the RGB cube
  - Obtain a colour hexagon by viewing the RGB cube along the diagonal from the white vertex to the origin



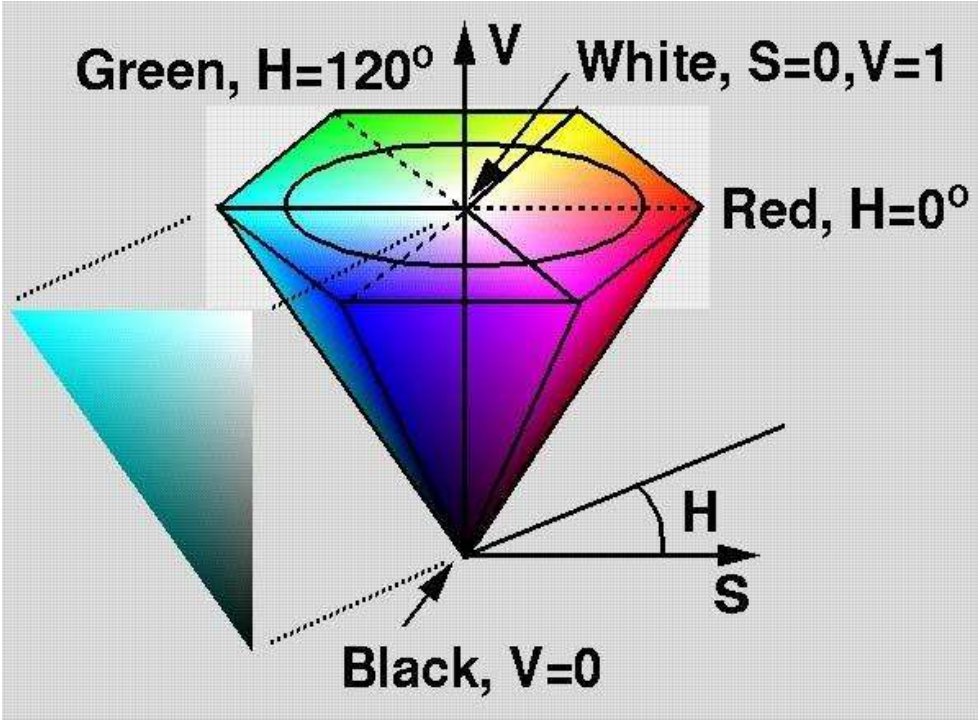
The HSV Colour Model

- The HSV hexcone
  - Hue is represented as an angle about the vertical axis ranging from 0 degree at red to 360 degree
  - Saturation parameter is used to designate the purity of a colour
  - Value is measured along a vertical axis through center of hexcone



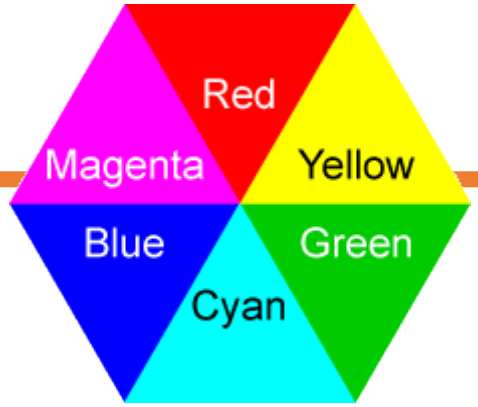


HSV Colour Model Hexcone

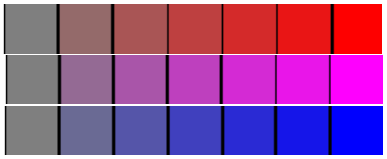


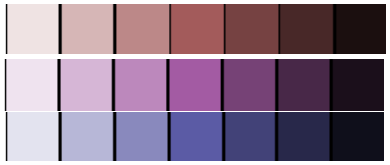
- Colour components:
  - Hue (H) ∈ [0°, 360°]
  - Saturation (S) ∈ [0, 1]
  - Value (V) ∈ [0, 1]

The HSV Colour Model

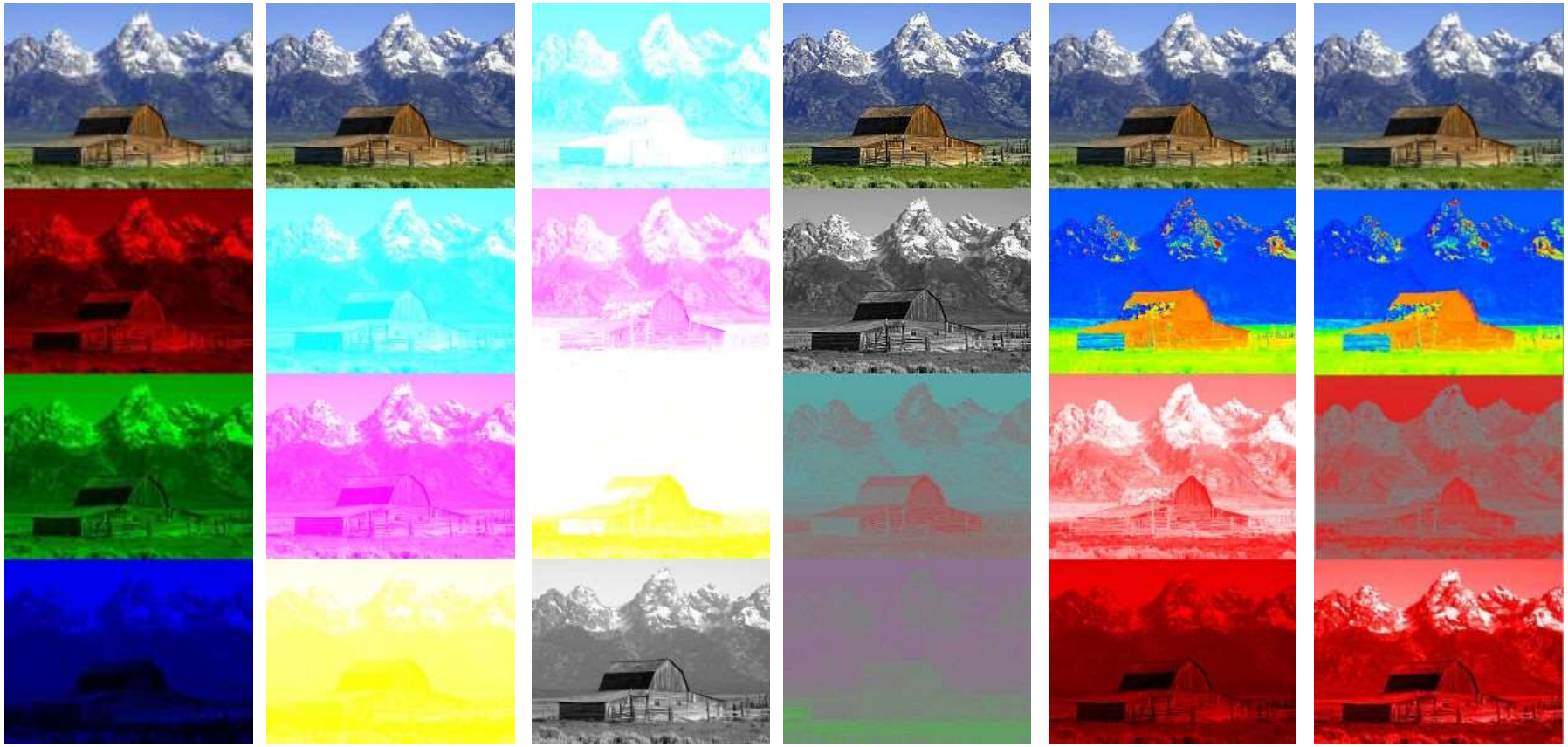


- Hue is the most obvious characteristic of a colour
- Chroma is the purity of a colour
  - High chroma colors look rich and full
  - Low chroma colors look dull and grayish
  - Sometimes chroma is called saturation
- Value is the lightness or darkness of a colour
  - Sometimes light colors are called tints, and
  - Dark colors are called shades





Comparison



RGB CMY CMYK YIQ HSV HSL

Colour Selection and Applications

- Graphical package provide colour capabilities in a way that aid users in making colour selections
  - For example, contain sliders and colour wheels for RGB components instead of numerical values

- Colour applications guidelines
  - Displaying blue pattern next to a red pattern can cause eye fatigue
    - Prevent by separating these colour or by using colours from one-half or less of the colour hexagon in the HSV model
  - Smaller number of colors produces a better looking display
  - Tints and shades tend to blend better than pure hues
  - Gray or complement of one of the foreground colour is usually best for background